

Juan Pablo Schamun

Senior Data Engineer & AI Specialist

San Carlos de Bariloche, Argentina

📞 (+54) 9 291 5714529

✉️ juanpsch@gmail.com

in www.linkedin.com/in/juanpsch/

📁 [juanpsch/portfolio](#)

Professional Summary

Senior Engineer with **10+ years of experience** bridging the gap between industrial operations and data technologies. Expert in **Data Engineering** (SQL, Oracle, ETL, Migrations) and Product Lifecycle Management (PLM). Currently finalizing a specialization in **Artificial Intelligence**, with practical implementation of ****RAG architectures****, Computer Vision, and NLP. Proven ability to lead technical teams, automate complex workflows using Python, and manage large-scale database migrations in critical environments (Nuclear Energy sector). Seeking to leverage deep data expertise and modern AI stack (PyTorch, Docker, LLMs) in a challenging remote role.

Technical Skills

Data Engineering	Advanced SQL (Complex Queries, Optimization), Oracle DB, Data Migration (ETL), Data Integrity Analysis.
AI & Machine Learning	Python (Pandas, NumPy, Scikit-Learn), PyTorch , Computer Vision (Vision Transformers), NLP, RAG, LLMs.
Development & DevOps	Docker (Containerization), Git , Linux, AWS (Familiarity), VBA/Macros.
Industrial Software	CATIA V5, ENOVIA V5, 3DEXperience, Smart Plant Instrumentation (SPI).
Languages	Spanish (Native), English (Advanced - CAE), German (Intermediate).

Professional Experience

Sept 2013 - Present	Lead PLM Data Engineer & Technical Coordinator , CNEA (<i>National Atomic Energy Commission</i>) - <i>Proyecto CAREM</i> , Bariloche, Argentina Led data management and digital infrastructure for the CAREM nuclear project, transitioning from traditional engineering to data-driven workflows. • AI & RAG Implementation: Architected a Retrieval-Augmented Generation (RAG) system to query thousands of confidential technical documents, deployed using Docker (customized RAG-FLOW) for scalability. • Database Migrations (ETL): Successfully executed critical database migrations for Smart Plant Instrumentation (SPI) across multiple versions (2009 → 2018) and ENOVIA updates, ensuring zero data loss. • Data Engineering & SQL: Designed and optimized complex SQL queries for Oracle databases and conducted rigorous data integrity checks. • Python Automation: Developed Python scripts (Pandas/NumPy) to automate data cleaning and report generation, reducing manual workload by 30%.
Jun 2012 - 2016	Independent Consultant , <i>Energy Efficiency & Data Analyst</i> , Bahía Blanca, Argentina Conducted technical assessments for renewable energy projects. Used data modeling to optimize energy consumption for industrial clients.

Nov 2008 **Contact Engineer**, *EXXONMOBIL*, Argentina
Jun 2009 Coordinated plant turnaround operations, ensuring adherence to strict technical procedures and minimizing downtime through precise planning.

Education

In Progress **Specialization in Artificial Intelligence**, *Universidad de Buenos Aires (UBA)*
(Thesis Phase) Thesis: RAG System for Confidential Document Management.
2012 **Master in Energy Technology for Sustainable Development**, *Universidad Politécnica de Valencia*, Valencia, Spain
2009 **Industrial Engineering**, *Universidad Nacional del Sur*, Bahía Blanca, Argentina

Key Projects & Portfolio

Computer Vision Project **Food Classifier ViT**: Fine-tuned Vision Transformers (ViT) using **PyTorch** to classify food images, demonstrating transfer learning and deep learning proficiency.
Thesis Project **RAG Architecture**: Architecting a production-ready Retrieval-Augmented Generation system using LangChain/RAG-FLOW and **Docker** to solve critical document querying for enterprise use.
Portfolio Link [juanpsch/portfolio](#) (Contains code for ViT, ML Scripts, and more.)